

Prototype Design Report

Project: 4-Stage Packed-Bed Water Filtration Column
Focus: Initial Prototype (V1) Testing, Failure Analysis, and Design Improvements

1. Overview

Before building the final filter, an initial prototype (V1) was constructed to test fluid flow, layer stability, and basic water clarification. The goal was to identify physical issues early and refine the design before running final data trials.

2. Initial Prototype Configuration (V1)

- **Body:** Single 500mL plastic bottle container.
- **Layer Setup (Top to Bottom):** Gravel → Fine Sand → Activated Carbon → Cotton Wad.
- **Outlet Design:** 1 cotton ball.

3. Testing & Observed Issues

Issue Observed	Root Cause	Impact on Performance
Cap Leaking / Pressure Buildup	Water pressure accumulated near the cap outlet.	Minor leakage around cap threads during initial pouring. Along with the cotton ball falling out
Cloudy Water Output (Black Dust)	Dry activated carbon had fine dust that was not washed out before testing.	High initial output turbidity (cloudy water) on the first run.
High Initial Water Loss	Dry sand and carbon soaked up water to fill internal air gaps.	Significant liquid loss during Run 1 (~% holdup).

4. Modifications for Final Design (V2)

1. **Redesigned Outlet Cap:** Expanded drainage area to allow smoother water flow without pressure buildup.
2. **Created a collection system:** Introduced a collection system for the water to make qualitative observations.
3. **Accounted for Media Saturation:** Recognized that Run 1 acts as a "wetting run," while Runs 2 and 3 deliver true volumetric output (~95% recovery).

5. Summary Comparison

Feature	Prototype (V1)	Final Design (V2)
Outlet Geometry	Single cotton-ball	Expanded drainage cap area added a mesh
Pre-Treatment	None (Dry media used directly)	Flushed with clean water prior to testing
Flow Rate Stability	Unstable / Backed up	Steady flow of 2.07mL/s
Output Clarity	Cloudy (Charcoal fines present)	Cleaner (charcoal was still found, just less)